

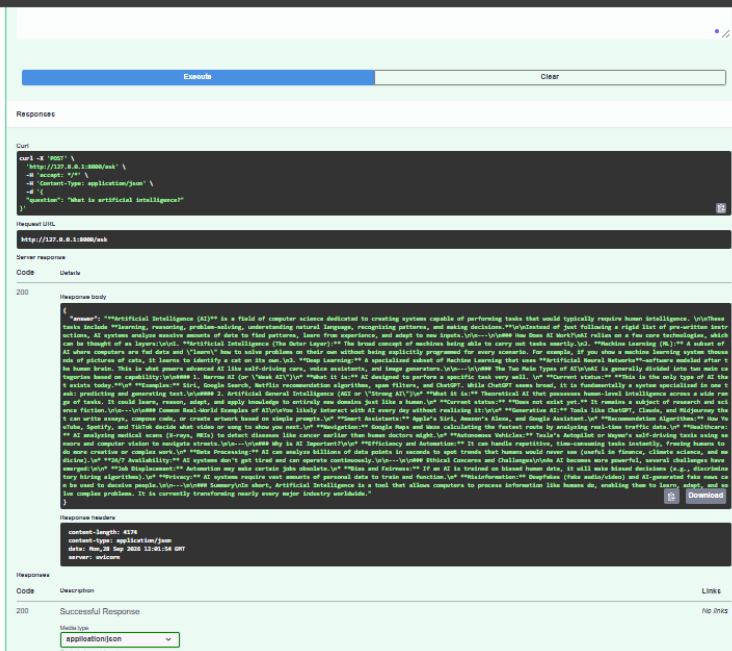
AI SOFTWARE DEVELOPMENT INTERNSHIP

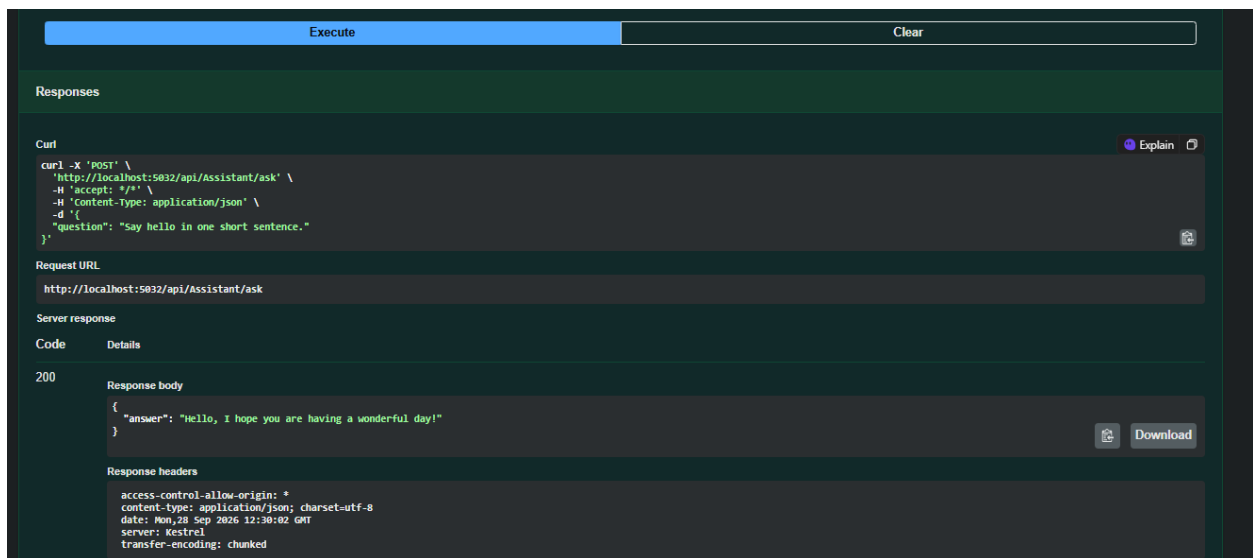
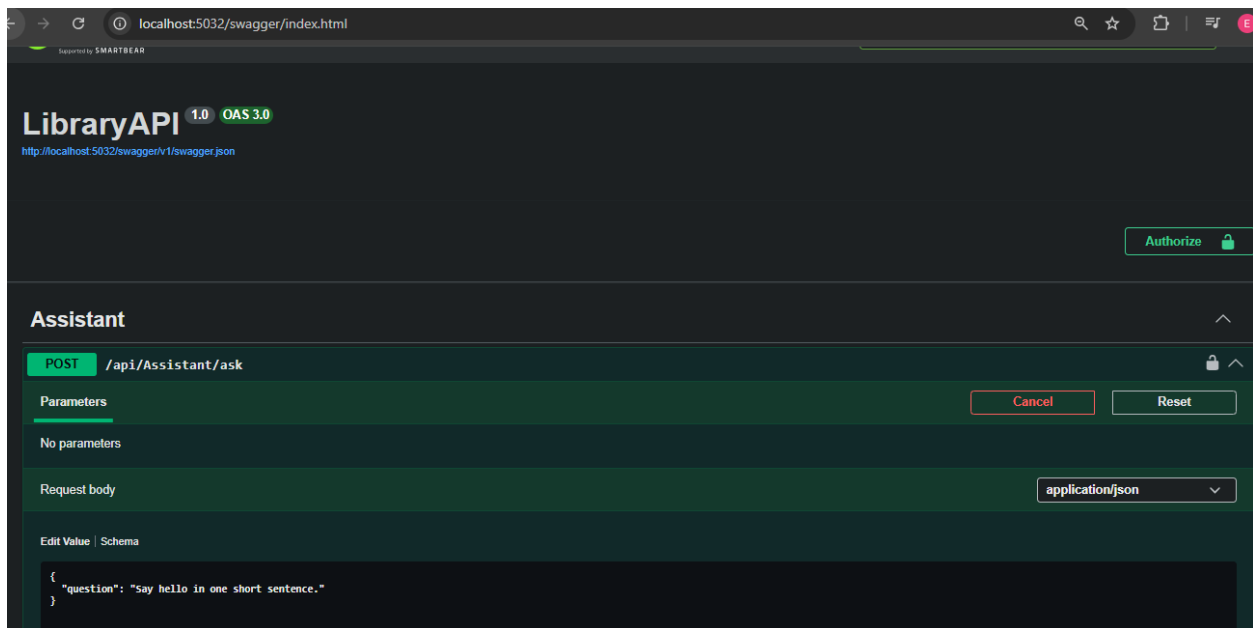
EMAN MASOOD (BSSE)

Week 6 — Detailed Task Sheet

LangChain & LCEL • Advanced Retrieval • Tool Calling Resilient .NET fi AI
Service Integration • End-to-End Streaming • Git Rebase

Outputs





localhost:5032/swagger/index.html

Assistant

POST /api/Assistant/ask

Parameters Cancel Reset

No parameters

Request body application/json

Edit Value | Schema

```
{
  "question": "Test AI service availability."
}
```

Execute Clear

Responses

localhost:5032/swagger/index.html

Execute Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:5032/api/Assistant/ask' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "question": "Test AI service availability."
  }'
```

Request URL

http://localhost:5032/api/Assistant/ask

Server response

Code	Details
503 Undocumented	Error: Service Unavailable Response body <pre>{ "message": "The AI assistant is temporarily unavailable. Please try again shortly." }</pre> Response headers <pre>access-control-allow-origin: * content-type: application/json; charset=utf-8 date: Mon, 28 Sep 2026 13:30:52 GMT server: Kestrel transfer-encoding: chunked</pre>

Responses

Code	Description
200	OK

```

        at System.Net.Http.HttpConnectionPool.SendWithVersionDetectionAndRetryAsync(HttpRequestMessage request, Boolean
        async, Boolean doRequestAuth, CancellationToken cancellationToken)
        at System.Net.Http.DiagnosticsHandler.SendAsyncCore(HttpRequestMessage request, Boolean async, CancellationToka
        n cancellationToken)
        at System.Net.Http.RedirectHandler.SendAsync(HttpRequestMessage request, Boolean async, CancellationToken cance
        llationToken)
        at System.Net.Http.SocketsHttpHandler.<SendAsync>g__CreateHandlerAndSendAsync|115_0(HttpRequestMessage request,
        CancellationToken cancellationToken)
        at Microsoft.Extensions.Http.Logging.LoggingHttpMessageHandler.<SendCoreAsync>g__Core|4_0(HttpRequestMessage re
        quest, Boolean useAsync, CancellationToken cancellationToken)
        [Retry] Attempt 1 after 2 seconds.
        info: System.Net.Http.HttpClient.IAiServiceClient.ClientHandler[100]
        Sending HTTP request POST http://127.0.0.1:8000/ask
        info: System.Net.Http.HttpClient.IAiServiceClient.ClientHandler[104]
        HTTP request failed after 2025.0932ms
        System.Net.Http.HttpRequestException: No connection could be made because the target machine actively refused it.
        (127.0.0.1:8000)
        ----> System.Net.Sockets.SocketException (10061): No connection could be made because the target machine actively
        refused it.
        at System.Net.Sockets.Socket.AwaitableSocketAsyncEventArgs.ThrowException(SocketError error, CancellationToken

```

```

        n cancellationToken)
        at System.Net.Http.RedirectHandler.SendAsync(HttpRequestMessage request, Boolean async, CancellationToken cance
        llationToken)
        at Microsoft.Extensions.Http.Logging.LoggingHttpMessageHandler.<SendCoreAsync>g__Core|4_0(HttpRequestMessage re
        quest, Boolean useAsync, CancellationToken cancellationToken)
        [Retry] Attempt 2 after 4 seconds.
        info: System.Net.Http.HttpClient.IAiServiceClient.LogicalHandler[100]
        Start processing HTTP request POST http://127.0.0.1:8000/ask
        info: System.Net.Http.HttpClient.IAiServiceClient.ClientHandler[100]
        Sending HTTP request POST http://127.0.0.1:8000/ask
        info: System.Net.Http.HttpClient.IAiServiceClient.ClientHandler[104]
        HTTP request failed after 2040.1147ms
        System.Net.Http.HttpRequestException: No connection could be made because the target machine actively refused it.
        (127.0.0.1:8000)
        ----> System.Net.Sockets.SocketException (10061): No connection could be made because the target machine actively
        refused it.
        at System.Net.Sockets.Socket.AwaitableSocketAsyncEventArgs.ThrowException(SocketError error, CancellationToken
        cancellationToken)

```

```

        at System.Net.Http.RedirectHandler.SendAsync(HttpRequestMessage request, Boolean async, CancellationToken cance
        llationToken)
        at Microsoft.Extensions.Http.Logging.LoggingHttpMessageHandler.<SendCoreAsync>g__Core|4_0(HttpRequestMessage re
        quest, Boolean useAsync, CancellationToken cancellationToken)
        [Circuit Breaker] Circuit opened for 30 seconds.
        [Retry] Attempt 1 after 2 seconds.
        info: System.Net.Http.HttpClient.IAiServiceClient.LogicalHandler[100]
        Start processing HTTP request POST http://127.0.0.1:8000/ask
        [Circuit Breaker] Circuit is half-open. Testing service.
        info: System.Net.Http.HttpClient.IAiServiceClient.ClientHandler[100]
        Sending HTTP request POST http://127.0.0.1:8000/ask
        info: System.Net.Http.HttpClient.IAiServiceClient.ClientHandler[104]
        HTTP request failed after 2032.4684ms
        System.Net.Http.HttpRequestException: No connection could be made because the target machine actively refused it.
        (127.0.0.1:8000)
        ----> System.Net.Sockets.SocketException (10061): No connection could be made because the target machine actively
        refused it.
        at System.Net.Sockets.Socket.AwaitableSocketAsyncEventArgs.ThrowException(SocketError error, CancellationToken
        cancellationToken)

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```

CancellationToken cancellationToken)
    at Microsoft.Extensions.Http.Logging.LoggingHttpMessageHandler.<SendCoreAsync>g__Core|4_0(HttpRequestMessage re
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    Sending HTTP request POST http://127.0.0.1:8000/ask
[Circuit Breaker] Circuit opened for 30 seconds.
info: System.Net.Http.HttpClient.IAIServiceClient.ClientHandler[104]
    HTTP request failed after 2049.2645ms
    System.Net.Http.HttpRequestException: No connection could be made because the target machine actively refused it.
(127.0.0.1:8000)
    ---> System.Net.Sockets.SocketException (10061): No connection could be made because the target machine actively
refused it.
    at System.Net.Sockets.Socket.AwaitableSocketAsyncEventArgs.ThrowException(SocketError error, CancellationTok
cancellationToken)

```

```

System.Net.Http.HttpRequestException: No connection could be made because the target machine actively refused it. (127.0.0.1:8000)
S D:\Week4\Project\LibraryAPI\ dotnet run
    at System.Net.Sockets.Socket.AwaitableSocketAsyncEventArgs.System.Threading.Tasks.Sources.IValueTaskSource.GetResult(Int16 token)
    at System.Net.Http.HttpConnectionPool.ConnectToTcpHostAsync(String host, Int32 port, HttpRequestMessage initialRequest, Boolean async, CancellationTok cancellationToken)
    --- End of inner exception stack trace ---
    at System.Net.Http.HttpConnectionPool.ConnectToTcpHostAsync(String host, Int32 port, HttpRequestMessage initialRequest, Boolean async, CancellationTok cancellationToken)
    at System.Net.Http.HttpConnectionPool.ConnectAsync(HttpRequestMessage request, Boolean async, CancellationTok cancellationToken)
    at System.Net.Http.HttpConnectionPool.CreateHttp11ConnectionAsync(HttpRequestMessage request, Boolean async, CancellationTok cancellationToken)
    at System.Net.Http.HttpConnectionPool.InjectNewHttp11ConnectionAsync(QueueItem queueItem)
    at System.Threading.Tasks.TaskCompletionSourceWithCancellation`1.WaitWithCancellationAsync(CancellationTok cancellationToken)
    at System.Net.Http.HttpConnectionPool.SendWithVersionDetectionAndRetryAsync(HttpRequestMessage request, Boolean async, Boolean doRequestAuth, CancellationTok cancellationToken)
    at System.Net.Http.DiagnosticsHandler.SendAsyncCore(HttpRequestMessage request, Boolean async, CancellationTok cancellationToken)
    at System.Net.Http.RedirectHandler.SendAsync(HttpRequestMessage request, Boolean async, CancellationTok cancellationToken)
    at Microsoft.Extensions.Http.Logging.LoggingHttpMessageHandler.<SendCoreAsync>g__Core|4_0(HttpRequestMessage request, Boolean useAsync, CancellationTok cancellationToken)
[Retry] Attempt 1 after 2 seconds.
info: System.Net.Http.HttpClient.IAIServiceClient.LogicalHandler[100]
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info: System.Net.Http.HttpClient.IAIServiceClient.LogicalHandler[100]
    Start processing HTTP request POST http://127.0.0.1:8000/ask
[Circuit Breaker] Circuit is half-open. Testing service.
info: System.Net.Http.HttpClient.IAIServiceClient.ClientHandler[100]
    Sending HTTP request POST http://127.0.0.1:8000/ask
info: System.Net.Http.HttpClient.IAIServiceClient.ClientHandler[104]
    HTTP request failed after 2028.6196ms
    System.Net.Http.HttpRequestException: No connection could be made because the target machine actively refused it. (127.0.0.1:8000)
    ---> System.Net.Sockets.SocketException (10061): No connection could be made because the target machine actively refused it.
    at System.Net.Sockets.Socket.AwaitableSocketAsyncEventArgs.ThrowException(SocketError error, CancellationTok cancellationToken)
    at System.Net.Sockets.Socket.AwaitableSocketAsyncEventArgs.System.Threading.Tasks.Sources.IValueTaskSource.GetResult(Int16 token)
    at System.Net.Http.HttpConnectionPool.ConnectToTcpHostAsync(String host, Int32 port, HttpRequestMessage initialRequest, Boolean async, CancellationTok cancellationToken)
    --- End of inner exception stack trace ---
    at System.Net.Http.HttpConnectionPool.ConnectToTcpHostAsync(String host, Int32 port, HttpRequestMessage initialRequest, Boolean async, CancellationTok cancellationToken)
    at System.Net.Http.HttpConnectionPool.ConnectAsync(HttpRequestMessage request, Boolean async, CancellationTok cancellationToken)
    at System.Net.Http.HttpConnectionPool.CreateHttp11ConnectionAsync(HttpRequestMessage request, Boolean async, CancellationTok cancellationToken)
    at System.Net.Http.HttpConnectionPool.InjectNewHttp11ConnectionAsync(QueueItem queueItem)
    at System.Threading.Tasks.TaskCompletionSourceWithCancellation`1.WaitWithCancellationAsync(CancellationTok cancellationToken)
    at System.Net.Http.HttpConnectionPool.SendWithVersionDetectionAndRetryAsync(HttpRequestMessage request, Boolean async, Boolean doRequestAuth, CancellationTok cancellationToken)
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    at System.Net.Http.RedirectHandler.SendAsync(HttpRequestMessage request, Boolean async, CancellationTok cancellationToken)
    at Microsoft.Extensions.Http.Logging.LoggingHttpMessageHandler.<SendCoreAsync>g__Core|4_0(HttpRequestMessage request, Boolean useAsync, CancellationTok cancellationToken)
[Circuit Breaker] Circuit opened for 30 seconds.
[Retry] Attempt 1 after 2 seconds.
info: System.Net.Http.HttpClient.IAIServiceClient.LogicalHandler[100]
    Start processing HTTP request POST http://127.0.0.1:8000/ask

```

PART E – RESILIENCE ANALYSIS

1. Why Immediate Retry Without Backoff Can Worsen a Struggling AI Service

If an AI service is already failing or overloaded, immediately sending repeated retry requests can increase the load on the service. This may cause more failures, higher resource usage, and slower recovery.

A retry storm can also occur when many clients retry at the same time.

To avoid this problem, exponential backoff is used. In this project, the retry delays increase between attempts:

Retry 1 → 2 seconds

Retry 2 → 4 seconds

Retry 3 → 8 seconds

This gives the AI service some time to recover before another request is sent.

2. Risk of Retrying a Request After Partial Success

A request may reach the AI provider successfully, but the response may fail to reach our application because of a network problem or timeout.

If the application automatically retries the same request, the AI provider may receive the request more than once.

This can cause:

- Duplicate processing
- Additional API costs for paid AI requests
- Duplicate side effects if the operation changes data or performs an action

Therefore, retries should mainly be used for transient failures. Requests that may have already been processed should be handled carefully. Where supported, idempotency keys can also help prevent duplicate processing.

3. Graceful Degradation

When the AI service was unavailable, the .NET API did not return a generic server error. Instead, it returned HTTP 503 with a clear message:

"The AI assistant is temporarily unavailable. Please try again shortly."

This allows the application to fail gracefully while the AI service is unavailable.

4. Circuit Breaker

The circuit breaker detected repeated failures and opened the circuit for 30 seconds.

The terminal output showed:

[Circuit Breaker] Circuit opened for 30 seconds.

After the cooldown period, the circuit entered the half-open state and tested the AI service again:

[Circuit Breaker] Circuit is half-open. Testing service.

Because the FastAPI service was still unavailable, the circuit opened again.

Conclusion

Part E successfully implements resilient .NET-to-AI-service communication using a typed HttpClient, retry policy with exponential backoff, circuit breaker, timeout handling, and graceful degradation.